GEOG5000M Internship Project: Footfall at Kirkgate Market with Leeds City Council

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1 Introduction

Since it's opening, the Kirkgate Market has undergone a number of transformations, and now comprises of both an covered indoor area and an outdoor area towards the rear, and is one of the largest covered markets in Europe (González and Waley, 2013). Originally built as an open market in the 1800's, it lies on the east side of Leeds city centre with the front side lined by Vicar Lane to the west. The market is owned by Leeds City Council (LCC) and hosts vendors selling a wide range of products including fish and meat, fruit and vegetables, clothing and flowers.

The following investigation was undertaken in conjunction with LCC, who provided the data used.

1.1 Motivation

Over the course of the introductory meeting with representatives from LCC in December 2017, we were made aware that beyond the previously used data concerning footfall in the centre of Leeds, there were also data available regarding the number of pedestrians passing in and out of Kirkgate market. Although this data had been collected over a number of years, it had predominantly been used for reporting and provided to market vendors; very little analysis had been undertaken with it. This therefore provided an opportunity to undertake novel work that would be of use to LCC.

In recent years, members of the city centre management and markets teams have undertaken measures to drive growth in footfall (and consequently business) at Kirkgate market, such as the installation of a number of street-food stalls and the hosting of an increasing number of events. One of the main focuses of such measures is to draw more young families to the market, such as those who typically visit Victoria Gate to the north.

This investigation therefore sought to make use of the new data source to explore the impact of such measures, and examine avenues for further growth. A particular avenue that was tested in this investigation was that of opening the market on Sundays.

1.2 Data

This investigation makes use of two separate data sources, both of which pertain to counts of pedestrians in different geographic locations around the city of Leeds, and how they vary over 2017.

The first of these data sources contains data regarding Kirkgate market (to be referred to as markets data). This data is collected by PFM Footfall Intelligence, and was provided by Steve Mason (Head of Markets, LCC), and is a record of the counts of pedestrians passing through each of the market entrances (shown in Figure 1). The data was initially aggregated to a weekly level; data with finer time-granularity was later provided (aggregated to a daily level). The data aggregates inflows and outflows, removing direction; it is not possible to glean whether pedestrians favour certain entrances for either entering or exiting the market, or whether this would be affected by any other factors. It is therefore important to clarify that any phenomena observed in the data will be treated as symmetric unless stated otherwise, i.e. any analyses referring to flows of pedestrians into the market will, by default, apply to flows of pedestrians out of the market.

The second data source contains data regarding Leeds city centre, and was provided by Springboard via John Ebo (to be referred to as springboard data). The data details the counts of pedestrians passing through different parts of the city (shown in Figure 2)

on an hourly basis. Prior to usage in this investigation, the data had undergone some preprocessing by Annabel Whipp.

Aside from the differing with regards to the geographical locations that they covered, the two datasets also differ in the method of collection. The springboard data is collected via a set of cameras; a camera is stationed at each of the locations shown in the figure, and counts the number of torsos passing through a fixed frame. The market data, on the other hand, is collected using lightgates stationed over each of the market entrances; this is the reason for the data not containing direction.

1.3 Aims

With the above in mind, two aims were settled upon. These were as follows:

1. Analyse the markets data to better understand the flows into and out of the market.

2. Build a predictive model to produce estimate hypothetical count values for the Sundays in 2017.

The following section of this report (Section 2) attempts to address the latter of the two. It covers the building of models, and their usage in estimating counts for Sundays at Kirkgate market if the market had been open on those days; this employs the springboard data as input variables for the models, and the response variable takes the form of the daily total count from the markets data. This is supplemented by Appendix A, which seeks to undertake some analysis of the markets data, identifying keys trends and positing some ideas for why these trends might occur, thus targeting the first of the aims.

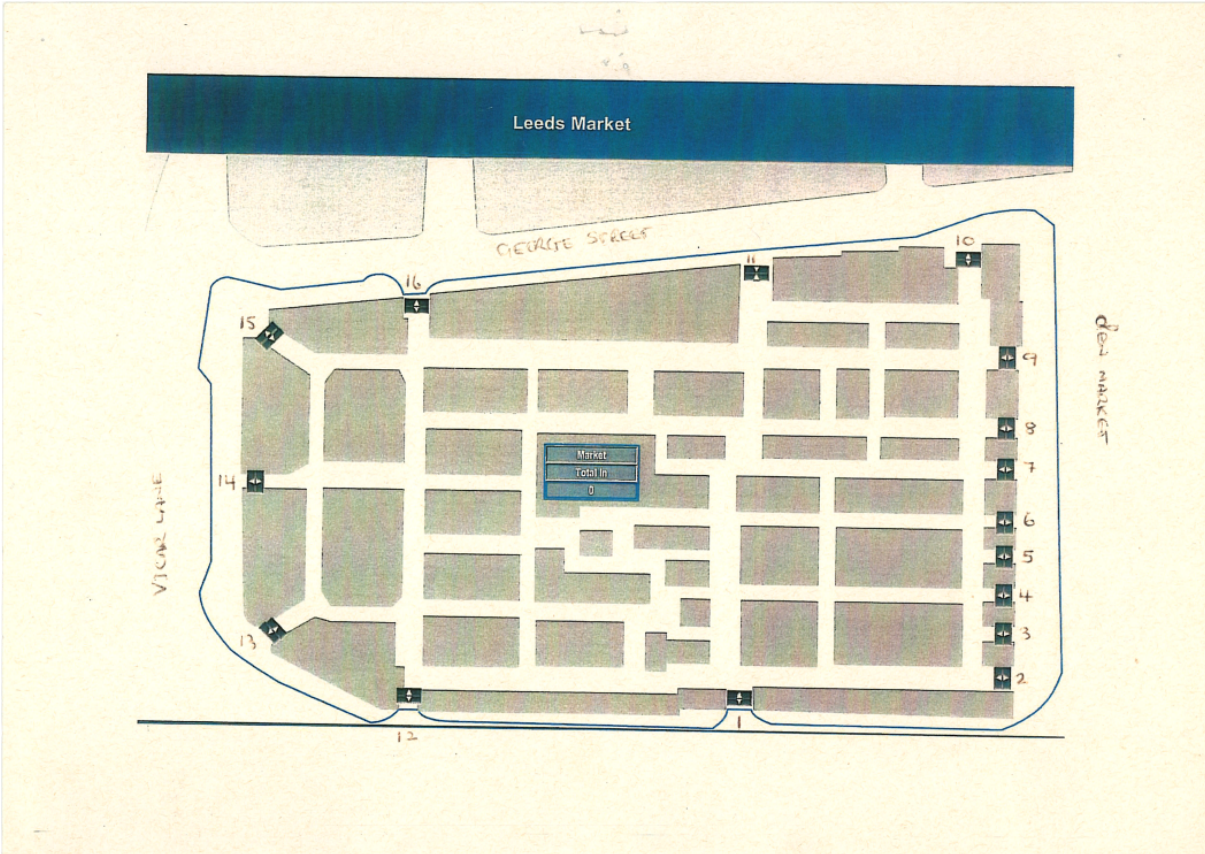


Figure 1: Floor-plan of Kirkgate market provided by Steve Mason.

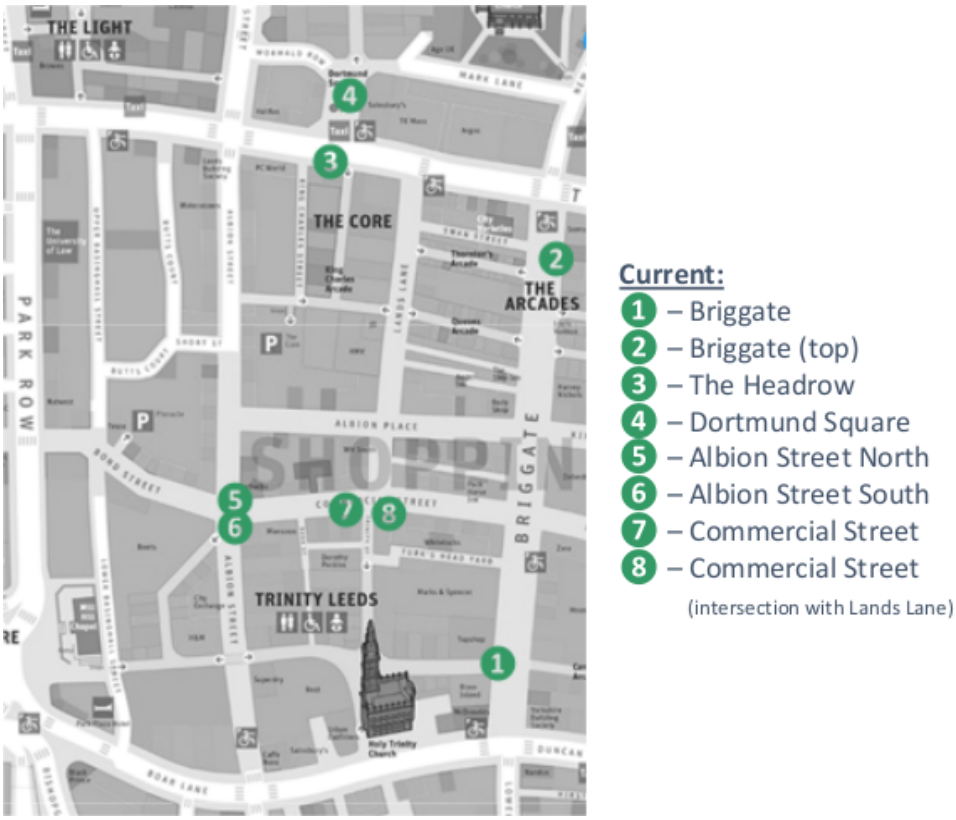


Figure 2: Locations of springboard footfall cameras in Leeds provided by Mark Durham.

2 Estimating Activity on Sundays

The aim of this section it to build a model that outputs total daily count values for Kirkgate market given city centre daily count values from the springboard data. This first requires aggregation of counts to the daily level for the springboard data for each of the camera sites. The two datasets are then joined on the `Date` variable, and models built.

2.1 Methods

In order to produce estimates of daily counts for Sundays, a range of different models were built using different model types (in `scikit-learn` (Buitinck et al., 2013))and different choices of input variables. The model types that were used for this investigation were:

- Linear regression (Isobe et al., 1990, Weisberg, 2005)
- Random forest regressor (Segal, 2004)
- Neural network regressor (Specht, 1991)

The power set of the input variables was produced (Devlin, 2012), thus accounting for every combination that could be fed into the models.

The markets data was then segmented into training dates and test dates, where the training dates would be used to train the models and validate them, and the test dates consisted of the Sundays in 2017 when Kirkgate market was closed; it is worth noting that the market is open on the two Sundays preceding Christmas.

Given that the dates for which our models were estimating did not have any actual values, it was important that the trained models were as accurate as possible, and that the best model was chosen to produce estimates. To achieve the former goal, k-fold cross-validation was applied (with $k = 3$ allowing for a validation set of $N \approx 100$ dates in each round), outputting average mean absolute error (MAE), root mean square error ($RMSE$) and R^2 for each model. The three measures are defined as follows:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - f_i|, \tag{1a}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - f_i)^2}, \tag{1b}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - f_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \tag{1c}$$

where y_i is the ith observation and f_i is the ith estimate for $i \in N$(Riley et al., 2006, Voss, 2013). Improvements in model performance are denoted by lower MAE, lower $RMSE$ and higher R^2.

2.2 Results

The aforementioned measures are plotted in Figure 3, allowing us to evaluate the performance of different families of models. Considering each of the plots, we are able to

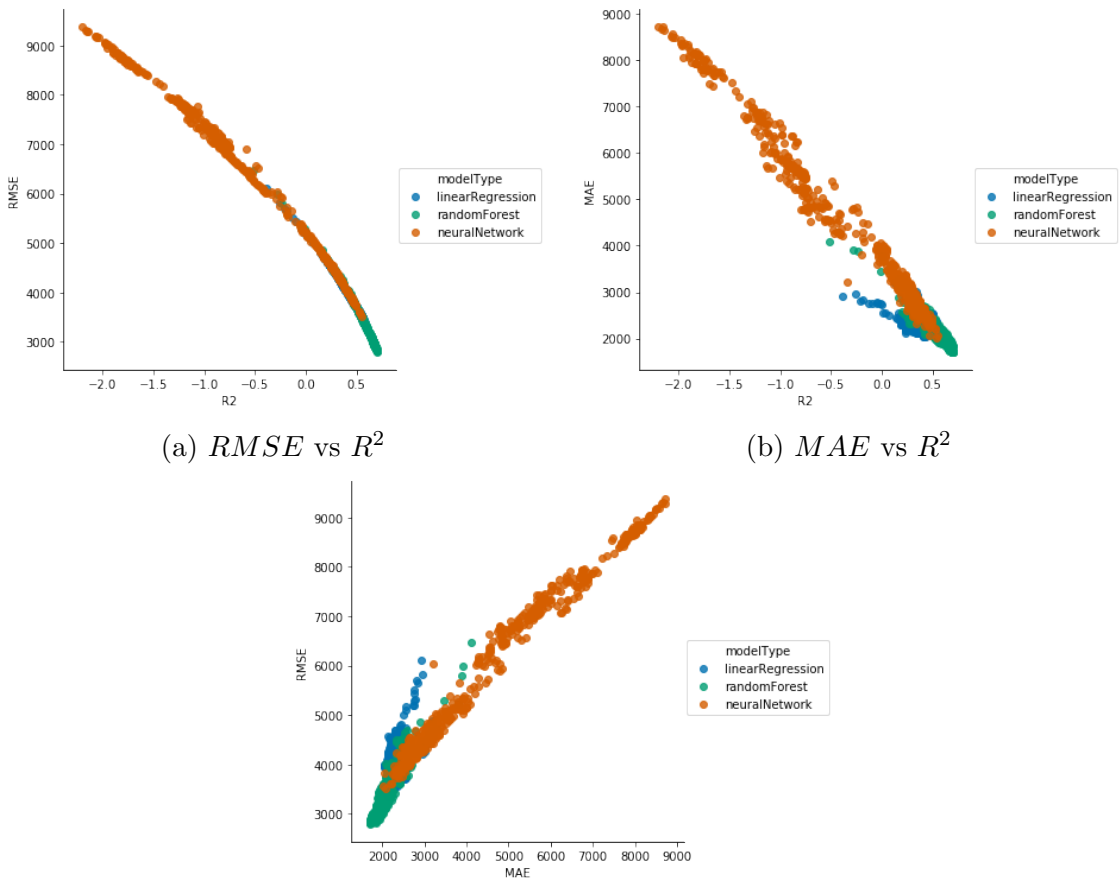
establish a general ordering of the families of models for each of the measures. The following ordering (of best to worst) appears to be common across each of the three measures:

1. Random forest
2. Linear regression
3. Neural network

Based on these measures, we can establish that random forest models perform best for our particular data set. In particular, the model

$$MarketTotal \sim RandomForest(AlbionStreetNorth, Briggate, BriggateAtMcDonalds, DortmundSquare, Headrow, Total) \quad (2)$$

performs best with $MAE = 1.71 \times 10^3$, $RMSE = 2.80 \times 10^3$ and $R^2 = 0.708$ (all given to 3 significant figures).



(a) $RMSE$ vs R^2

(b) MAE vs R^2

(c) $RMSE$ vs MAE

Figure 3: Evaluation of model performance by model family.

We proceed to produce estimates of total counts for the market on Sundays using this model, which are shown in Figure 4. The figure shows that, for the majority of weeks in the year, the estimated counts for Sundays are less that 25% of the values for the other days of the week.

It is perhaps more worthwhile comparing the estimates with the actual values for the last two Sundays of the year (for which the market is open). When considering these weeks as a benchmark, many of the estimates for October, November and December are comparable. This may suggest that there is some value to opening Kirkgate market on Sundays earlier in the Winter period. Acknowledgement should be given, however, to the MAE and $RMSE$ values obtained when training the model; these are significant in the context of the estimated values for Sundays. The estimated values are typically less than $10,000$, and as such the aforementioned MAE and $RMSE$ values represent errors upwards of 10%.

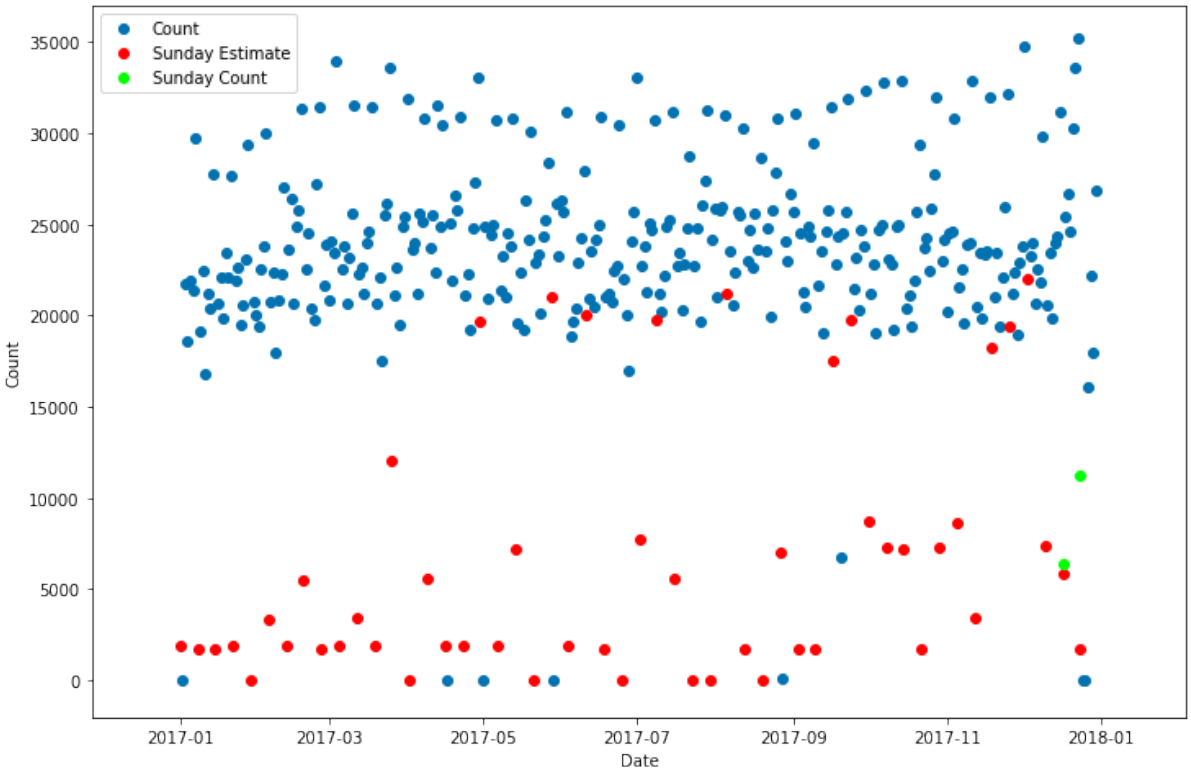


Figure 4: Estimations for the total counts across all entrances.

3 Conclusion

This investigation has sought to achieve two goals. The first of these was an exploratory analysis of the data provided regarding weekly counts of pedestrians passing through the entrance of Kirkgate market, which is covered in Appendix A. The second goal was to construct models to estimate hypothetical counts through the market entrances on Sundays in 2017, given that the market is typically closed on Sundays. It was found that this was best achieved using random forest models based on a selection of evaluative measures. The chosen model was then used to produce estimates.

Future Work

Given the investigation carried out, a number of suggestions for future work are suggested:

- A more involved analysis of the market data, looking to separate the trends associated with seasonality and the trends associated with events.
- Given that the input variables for the models are highly correlated (see Figure 5), we may consider performing a PCA; the hope is that this will aid the performance of the linear regression models in particular, as these models are susceptible to performance-loss when handling data in which there is multicollinearity.
- The use of other data sources in models; in particular, the inclusion of events would be of use.
- Treatment of outliers, particularly when considering data for modelling with linear regression.

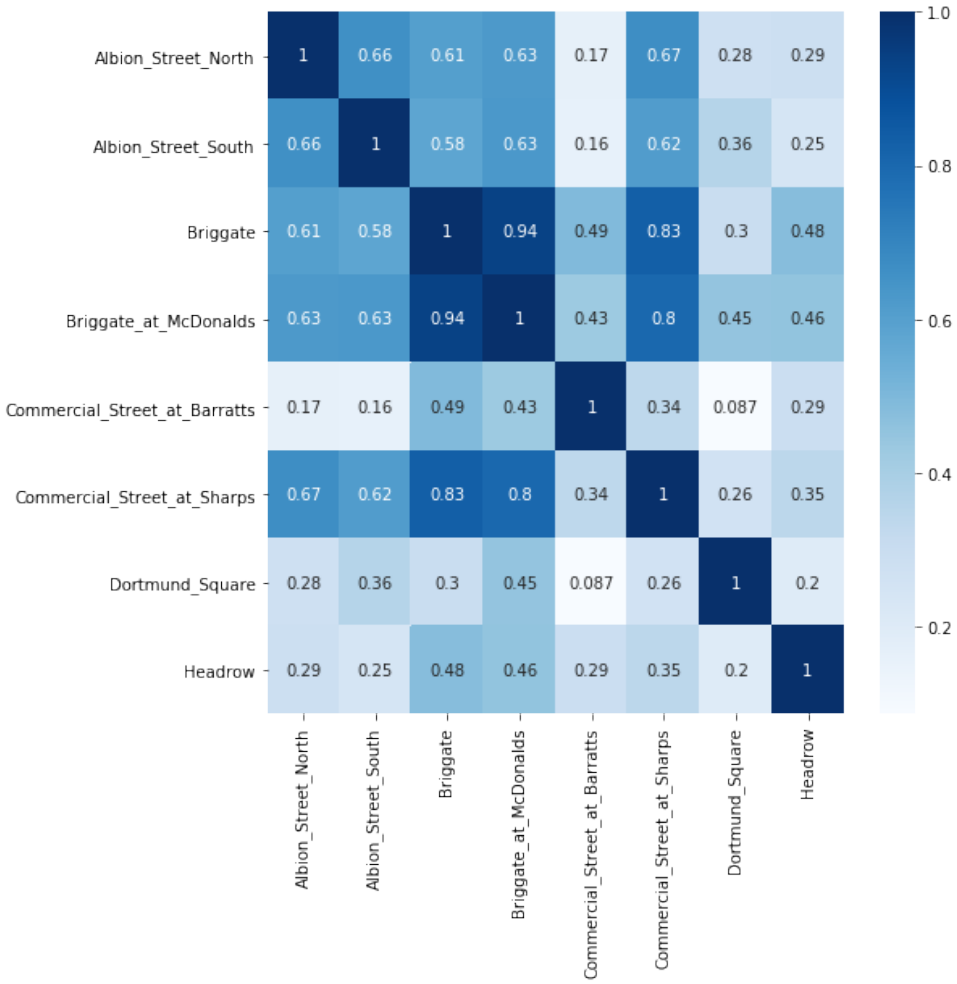


Figure 5: Heat-map of correlations between footfall counts at each of the camera locations.

4 Reflective Annex

There are two salient points that merit consideration when reflecting on my experience of undertaking the internhip with Leeds City Council. The first of these is the interpersonal relationship that was built over the course of the internship. Despite a minor delay in the commencement of the internship, I was able to spend all of my required working hours in the LCC offices. This was very useful as it allowed for the ability to immediately communicate with the necessary contacts. Whilst this granted me a sense of integration, I was also given a great deal of autonomy with regards to the direction in which I proceeded with the project — something that was appreciated. Furthermore, I felt that my skills and opinions were valued, and that there was a genuine interest in fostering an ongoing relationship for the remainder of the PhD process.

The other point worth noting is that of data limitations. The prior expectations of the markets data were perhaps too high — the hope was that the data would contain the direction in which pedestrians passed through each entrance, and that the counts would be recorded at an hourly level. In reality the data was aggregated at the daily/weekly level, and the data aggregated inflows and outflows, removing direction. It was therefore not possible to glean whether pedestrians favoured certain entrances for either entering or exiting the market, or whether this would be affected by any other factors. Furthermore, this made the data inappropriate for undertaking the original aim of producing a dynamically calibrated agent-based model (which would have required data at finer time-granularity). Whilst there were some discussions of trying to achieve this by simulating the hourly flows, the aims were ultimately modified to those now found in Section 1.3.

The modification of the project aims has also resulted in the use and development of a different set of technical skills. Whilst the original aims sought to foster a practical understanding of how to develop an agent-based model, and how one could be used in conjunction with data assimilation methods, the new aims (in particular the estimation element covered in Section 2) allowed me to apply my recently acquired knowledge of machine learning algorithms. Of particular use was the idea of cross-validation which was covered whilst attending the CDT module in Liverpool.

A Trends in Market Activity

This Section will go on to document the analysis that has been undertaken on the markets data, and the trends that this analysis has yielded. This will be split into three themes:

1. Section A.1 will focus on how counts vary across the different entrances.
2. Section A.2 will focus on how counts vary across the different sides.
3. Section A.3 will focus on how counts vary based on whether events are taking place at Kirkgate market.

A.1 Variation by Entrance

When first considering the variation in counts by entrance, the values for all of the entrances over the course of 2017 were plotted on one Figure (as seen in Figure 6). This plot contained far too much information to discern any noticeable trends; however, it was clear that the counts for entrances 3, 4 and 5 were 0 for all dates in the time-frame. A conversation with Steve Mason regarding this issue revealed that entrances 3,4 and 5 no longer have sensors on them as they were closed off as part of the refurbishment works in 2016 and not reopened. Beyond this, it is interesting to observe the variation between different entrances; it appears that counts through entrances 13 and 15 are consistently higher than the other entrances. This is unsurprising, given the location of entrances 13 and 15 — these are two of the three front-facing entrances on Vicar Lane.

To start looking at the variations by entrance, the entrances were divided up based on which side of the market they are on — north, west, south or east — as shown in Figure 1.

The entrances on the west side are entrances 13, 14 and 15. The variation in counts for each of these entrances can be seen in Figure 7. As previously mentioned, it is to be expected that the entrances on the west side of the market receive greater footfall counts — and indeed this is what we observe in the case of entrances 13 and 15. Entrance 14, however, garners significantly less footfall than the other two front-facing entrances, often around a quarter of the counts seen at the other two entrances. This is likely because the majority of pedestrians entering and exiting the market on that side are originally travelling along the length of Vicar Lane; anyone travelling north to south and intending to enter the market would enter through entrance 15, and anyone travelling south to north and intending to enter would enter through entrance 13.

The entrances on the north side are entrances 10, 11 and 16. The variation in counts for each of these entrances can be seen in Figure 8. Here, we can see that entrance 16 consistently receives between three and four times more footfall than the other two entrances. Entrance 16 is the entrance on the north side that is closest to Vicar Lane. It is, therefore, likely that the increase in counts through this entrance is a result of its proximity to the front of the market: in a similar fashion to entrance 15, pedestrians walking from north to south along Vicar Lane with the intention of reaching the back of the market may use entrance 16.

The entrances on the south side are entrances 1 and 12; these lead out onto New York Street and its intersection with Kirkgate. The variation in counts for each of these entrances can be seen in Figure 9. Of the two entrances, entrance 12 receives greater footfall counts. Once again, this is likely due to it being closer to the front side of the market and Vicar Lane.

The remaining entrances on the east side of Kirkgate market lead out to the open outdoor market, the market car-parks and the bus station. The variation in counts for each of these entrances can be seen in Figure 10, which shows that footfall through the these entrances varies substantially across the various entrances. The counts of pedestrians passing through entrances 6 and 9 are higher than those for other entrances; this may be related to the layout of the back section of the market, whereby it is split into two areas with the northern area acting more as a traditional market and the southern area acting more as a food court. Entrances 9 and 6 therefore appear to function as the primary entrances for each of these areas respectively.

A.2 Variation by Side

Let us consider how footfall counts vary across each of the sides of the market. This is achieved by aggregating the counts from each entrance into four new columns: north entrances, south entrances, west entrances, and east entrances.

Figure 11 shows that the entrances can be split into two groups: the first group contains the west and east side entrances, and the second group contains the north and south side entrances. The former groups receive higher footfall counts throughout the year, whilst the latter groups receive lower footfall counts.

The higher counts at the west and east side entrances are a result of two previously mentioned effects: high footfall through the west side entrances is a consequence of this being the front of the market, facing Vicar Lane and the city centre; high footfall through the east side entrances is likely a consequence of people using them to either leave to or enter from the car park and bus station.

Each of the collections of the east and west side entrances receive comparable counts of footfall despite a disparity in the number of entrances (3 on the west side and 5 on the east side). This is accounted for by the average for the east side entrances being lower over the course of the year.

A.3 Variation with Events

In order to understand the impact of events on the number of people passing through the entrances of Kirkgate market, the daily aggregated market data was used instead of the weekly aggregated data, as events line up with specific dates in the data. Using this data, distributions were produced of the total counts in a day when there were events and when there were no events (Figure 12). This figure shows that the distributions for each case is somewhat bimodal, with the lower mode lying around 25000 and the upper mode lying around 30000. The distinctions between the two are:

1. In the case of the distribution for days with events, the upper mode is more heavily weighted - approximately 3 times more than that of the distribution for no events.

2. The distribution for no events has a longer lower-side tail, indicating that events results in a lower base count.

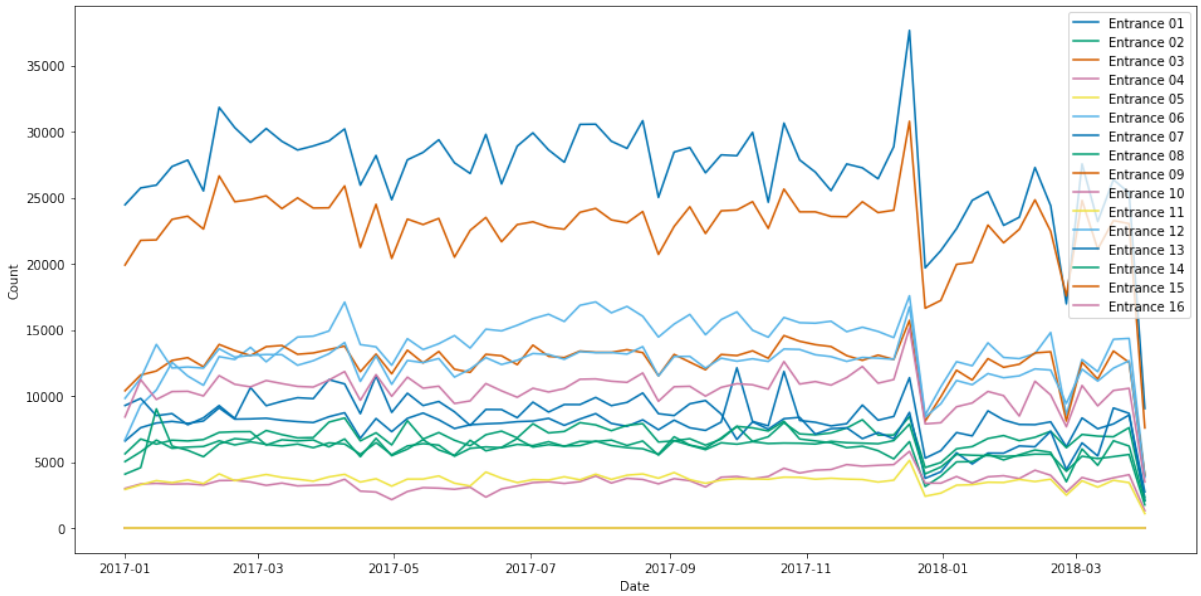


Figure 6: Counts detected through all entrances of Kirkgate market.

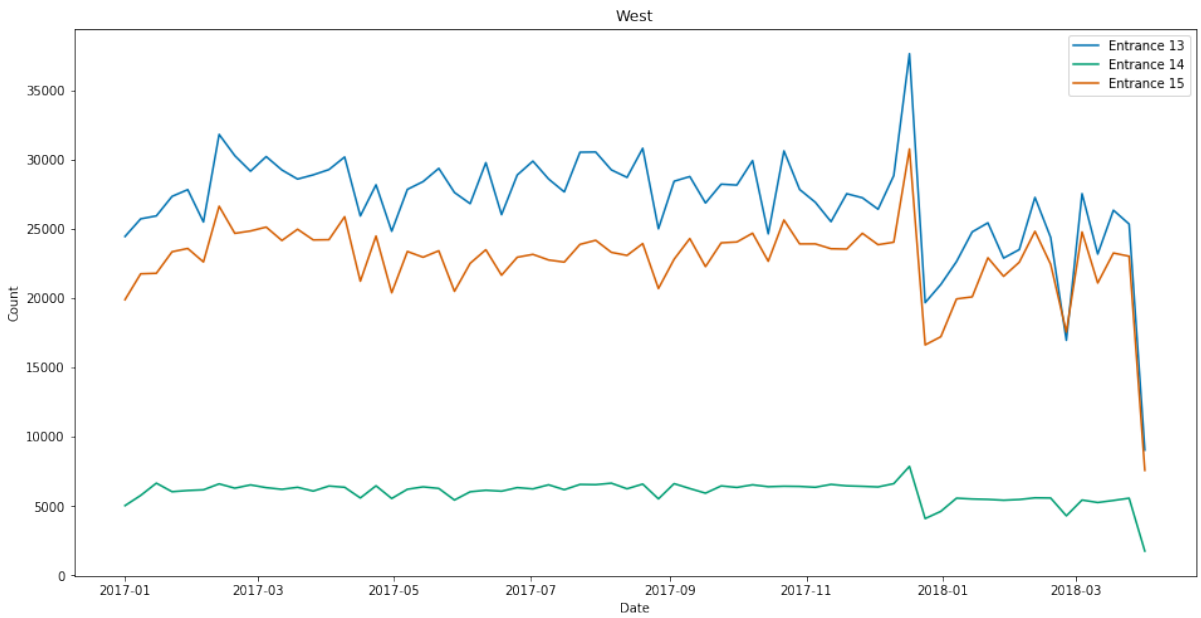


Figure 7: Counts detected through entrances on the west side of Kirkgate market.

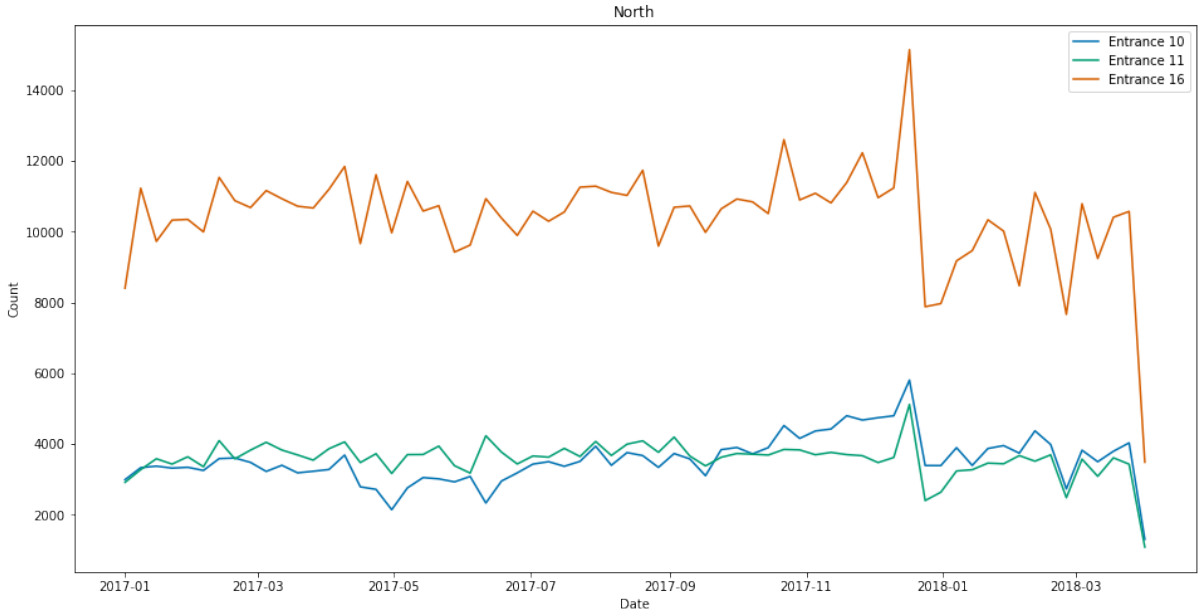


Figure 8: Counts detected through entrances on the north side of Kirkgate market.

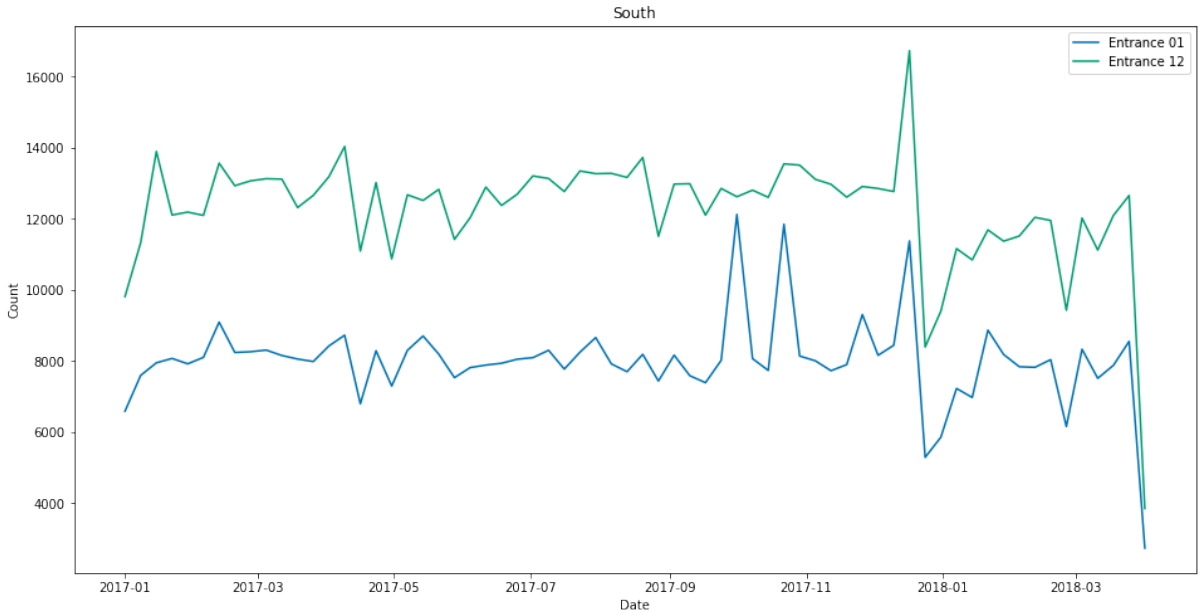


Figure 9: Counts detected through entrances on the south side of Kirkgate market.

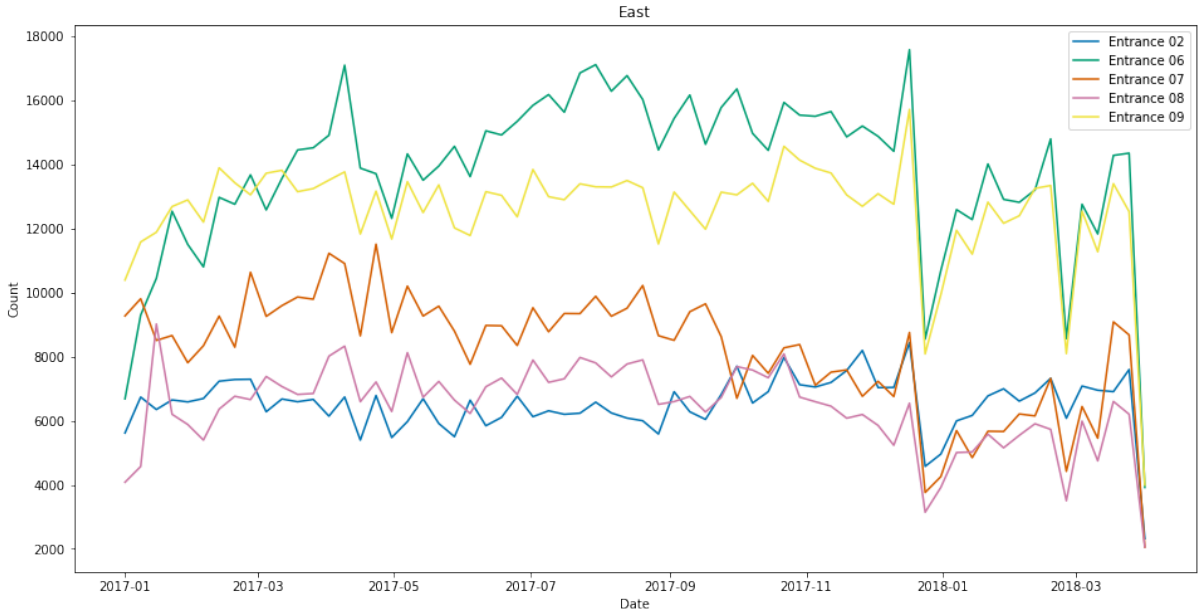


Figure 10: Counts detected through entrances on the east side of Kirkgate market.

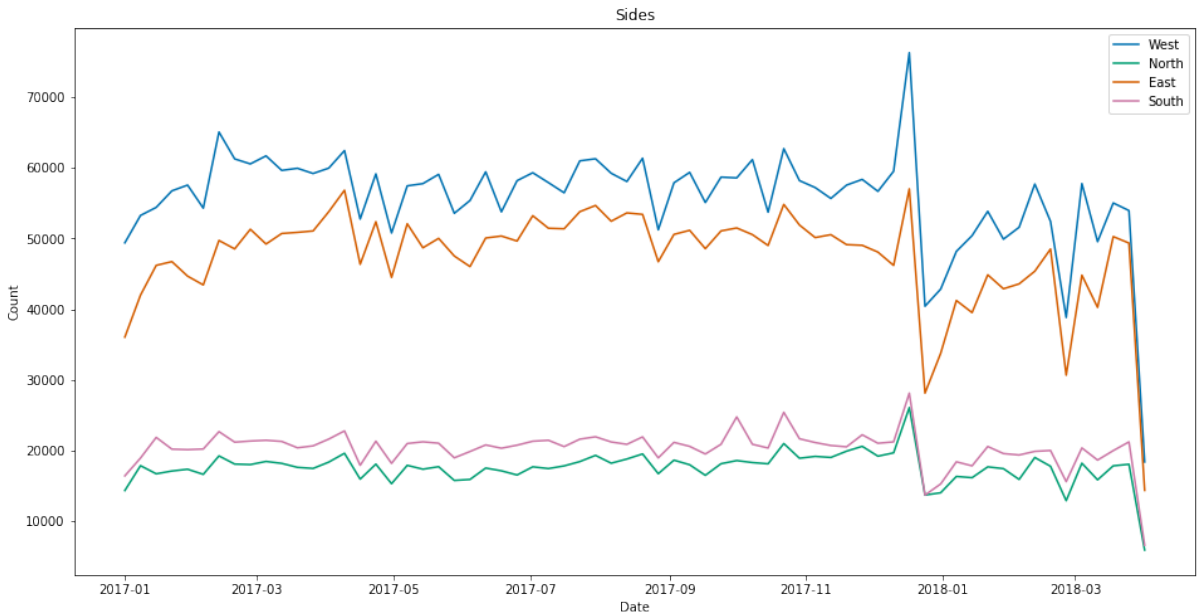


Figure 11: Variation of counts by side of Kirkgate market.

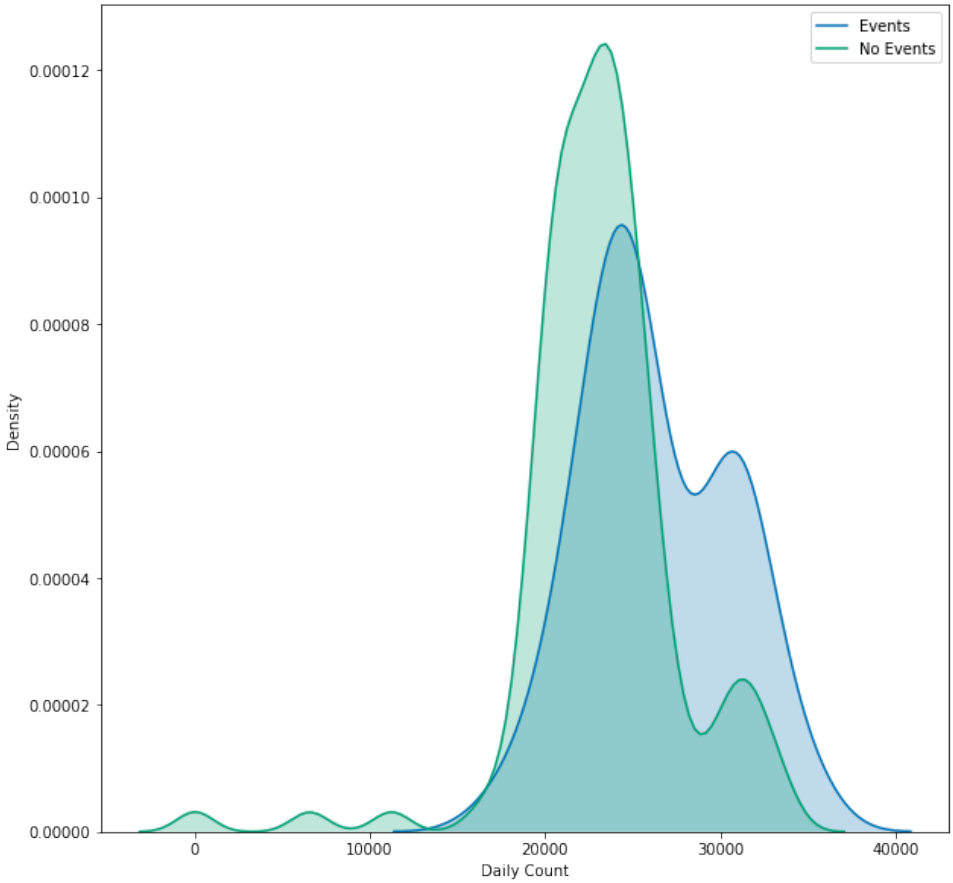


Figure 12: Distribution of counts based on events.

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